Generalized Multimodal Foundation Model

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Abstract

Making prediction with multimodal data is widely used in diverse scenarios. Existing multimodal fusion models, once deployed, can only handle predefined modalities (e.g., vision, text and audio) and single tasks, making it difficult to quickly adapt to new downstream applications. Therefore, a natural yet rather aggressive question arises, whether there exists a general multimodal fusion model that can be applied to arbitrary modality combinations and arbitrary prediction tasks. We argue that a unified multimodal fusion model should not depend on specific modalities and instead encode transferable patterns of multimodal correlation. To this end, we propose a simple and effective learning paradigm based on training over the generation of large-scale synthetic multimodal datasets with diverse causal structures that formally characterize the generative processes of multimodal data in real world. Building on this framework, we propose the generalized multimodal foundation model, a unified foundation model for generalized multimodal learning. By constructing large-scale synthetic multimodal datasets with diverse correlation patterns, our model encodes transferable multimodal correlations during training and activates appropriate associations through in-context examples during inference. Extensive experiments on 18 real-world datasets spanning 12 modalities and 11 prediction tasks demonstrate that our model achieves competitive performance with specialized models without task-specific adaptation.

1 Introduction

Making prediction with multimodal data focuses on exploiting synergy across modalities, enabling models to capture richer information and make better predictions [1, 2, 3]. By modeling interactions among heterogeneous modalities, multimodal learning has become an important paradigm for understanding complex real-world data [4]. However, existing multimodal fusion models are predominantly developed under a task-specific paradigm, where they are designed for predefined modalities and one specific prediction objective [5, 6, 7]. Consequently, these models mainly learn fusion strategies tailored to specific scenarios, making it difficult to capture generalized multimodal correlation patterns beyond predefined modality combinations and task settings.

The recent success of foundation models demonstrates that large-scale pretraining over diverse data distributions enables models to learn transferable patterns and acquire generalizable capabilities across diverse tasks and domains [8, 9, 10, 11, 12, 13]. Following this paradigm, multimodal foundation models have emerged to integrate heterogeneous information sources and advance multimodal understanding and prediction [14, 15, 16]. However, existing models remain largely confined to a limited modality space, focusing on several well-established modalities such as vision, language, and audio while leaving many other heterogeneous modalities largely unexplored. Such modality-specific designs hinder their ability to capture the complex correlations of generalized modalities that naturally arise in real-world scenarios. Accordingly, a fundamental and aggressive question arises:

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can we develop a multimodal foundation model that can rapidly adapt to arbitrary modality combinations and prediction tasks?

The first challenge of achieving such a generalized multimodal foundation model is the requirement of a large-scale and diverse collection of multimodal datasets that sufficiently cover the characteristics of real-world data. However, constructing such a collection using real datasets is inherently challenging. Existing multimodal datasets are highly concentrated on limited modality combinations [17, 18], while many other modalities remain scarce due to domain-specific acquisition processes and limited data availability. Moreover, collecting, annotating, and aligning large-scale datasets across diverse domains introduces substantial cost and practical difficulties. These limitations make it difficult to obtain comprehensive training data that cover diverse modality combinations and multimodal correlation patterns. Another challenge lies in the model design, namely how to develop a unified architecture that can adapt to future unseen prediction tasks. Existing multimodal models are predominantly optimized for predefined modalities and task objectives [15, 19], resulting in architectures that are tightly coupled with specific scenarios. Such designs limit their ability to transfer learned multimodal correlation patterns to new tasks and modality combinations.

To address these challenges, in this paper, we propose a new learning paradigm for generalized multimodal learning based on large-scale synthetic multimodal data. Instead of relying on real multimodal data with limited modality coverage, we construct synthetic multimodal data with Structural Multimodal Causal Model (SMCM), where complex multimodal correlations naturally emerge through the underlying data generation processes. By training over a broad collection of synthetic multimodal data, our model learns transferable multimodal correlation patterns beyond specific modality combinations and datasets. Building upon this paradigm, we develop the Generalized Multimodal Foundation Model by extending a context-based tabular foundation model, TabPFN [9], which provides strong capabilities in modeling complex dependencies from contextual information, toward generalized multimodal prediction. Through a unified token generation mechanism, the model maps heterogeneous modalities into a unified token space and enables multimodal in-context learning over diverse correlation patterns. During inference, the model leverages contextual examples from real datasets to activate task-relevant multimodal correlations and perform prediction without additional parameter updates. Extensive experiments across diverse real-world datasets demonstrate that our model achieves competitive performance with specialized multimodal fusion models while substantially improving generalization beyond predefined scenarios. Our contributions are summarized as follows:

- We introduce a new learning paradigm for generalized multimodal learning by constructing large-scale synthetic datasets with diverse SMCMs, enabling models to learn the general multimodal correlation patterns.
- We propose the Generalized Multimodal Foundation Model, a unified foundation model that adapts to diverse modality combinations and prediction tasks through multimodal in-context learning.
- Extensive experiments on 18 real-world datasets demonstrate that our generalized model achieves competitive performance with specialized models without task-specific adaptation.

2 Related Work

Multimodal Fusion. Multimodal fusion aims to integrate heterogeneous information from multiple modalities to learn more comprehensive representations [20, 21]. Early research mainly explores feature-level and decision-level fusion strategies. Feature-level fusion combines modality representations before prediction [22, 23], while decision-level fusion aggregates modality-specific predictions, providing modularity but limited cross-modal interaction [24, 6]. With the development of deep learning, intermediate fusion methods introduce modality-specific encoders and interaction modules to jointly learn modality representations and cross-modal correlations [2]. Recently, Transformer-based architectures have become the dominant paradigm due to their ability to model complex dependencies through attention mechanisms [18, 25, 26]. Existing approaches typically follow either single-stream architectures, which project multiple modalities into a shared representation space, or multi-stream architectures, which preserve modality-specific encoders and perform cross-attention fusion [27]. Furthermore, cross-attention mechanisms have been widely adopted to explicitly capture interactions between heterogeneous modalities and dynamically integrate complementary information [28, 29].

Despite their effectiveness on specific tasks, these methods usually assume predefined modality combinations and require task-specific architectures, limiting their generalization to diverse and unseen multimodal scenarios.

Multimodal Foundation Models. Recent advances in foundation models have shifted multimodal learning toward large-scale pretraining and transferable representations [30, 16]. Early multimodal foundation models mainly focus on learning aligned representations between vision and language, with representative works such as CLIP [31] and ALIGN [32]. Building upon these advances, multimodal large language models (MLLMs) further integrate visual encoders with large language models to enable multimodal understanding and reasoning, including Flamingo [33], BLIP-2 [34], and LLaVA [35]. More recent studies have explored unified multimodal architectures by mapping different modalities into shared token spaces and leveraging autoregressive modeling, such as Chameleon [36], Emu3 [15], and Janus [37]. However, existing multimodal foundation models are still primarily designed for limited modality combinations and often rely on extensive modality-specific pretraining, making it challenging to achieve generalization across arbitrary modalities and task settings.

3 Method

3.1 Preliminaries

Existing multimodal fusion methods typically follow a task-specific learning paradigm. Let $\mathcal{M}$ denote a generalized modality space encompassing all potentially available modalities. A real-world multimodal dataset $\mathcal{D}^{\text{real}}$ is associated with an available modality subset $\mathcal{S} \subseteq \mathcal{M}$ and a prediction objective. Given $\mathcal{D}^{\text{real}}$, existing fusion methods optimize a dedicated model $p_{\theta_{\mathcal{D}}}$ to capture the cross-modal correlations relevant to that particular dataset. Consequently, the learned model is inherently tied to its modality configuration and prediction objective, and typically needs to be re-optimized when applied to a new multimodal setting.

In contrast, our goal is to learn a generalized multimodal foundation model p_θ with a single set of parameters shared across diverse multimodal datasets and prediction tasks. Rather than fitting dataset-specific fusion functions, the model is expected to capture general multimodal correlation patterns that can be reused across different modality combinations. Once trained, p_θ can be directly applied to unseen real-world multimodal datasets without task-specific parameter updates.

3.2 Generation of Synthetic Multimodal Data

Training a generalized multimodal foundation model requires exposure to a large and diverse collection of multimodal datasets. However, constructing such a collection from real-world data is inherently difficult. Existing datasets cover only a limited range of modality combinations, while acquiring, aligning, and annotating heterogeneous modalities across different domains is often costly and domain dependent. As a result, real-world datasets alone provide insufficient coverage of the diverse multimodal correlation patterns required for generalized multimodal learning. To overcome this limitation, we construct synthetic multimodal datasets with controllable modality combinations and correlation structures, providing a broad training distribution for learning general multimodal patterns.

Figure 1 illustrates the motivation and overview of our synthetic multimodal data generation process. As shown in Figure 1(a), real-world data are commonly collected by multiple sensors observing the same underlying object or state, where different sensing processes expose different aspects of the underlying information. Inspired by this observation, Figure 1(b) depicts the basic principle adopted in our synthetic construction: a shared root representation R is transformed through distinct generation mechanisms f_m to produce heterogeneous modality representations M_m. This formulation naturally induces cross-modal correlation through a common underlying source. However, a direct transformation from R to each modality provides limited control over how information is shared among modalities or retained within individual modalities, making it difficult to systematically generate datasets with diverse forms of multimodal correlation.

To enable controllable construction of multimodal data with diverse correlation patterns, we propose a Structural Multimodal Causal Model (SMCM), as illustrated in Figure 1(c). An SMCM characterizes the multimodal generation process through a structured causal graph consisting of a root node, a Correlation Control Module (CCM), and multiple modality nodes. The CCM introduces shared nodes

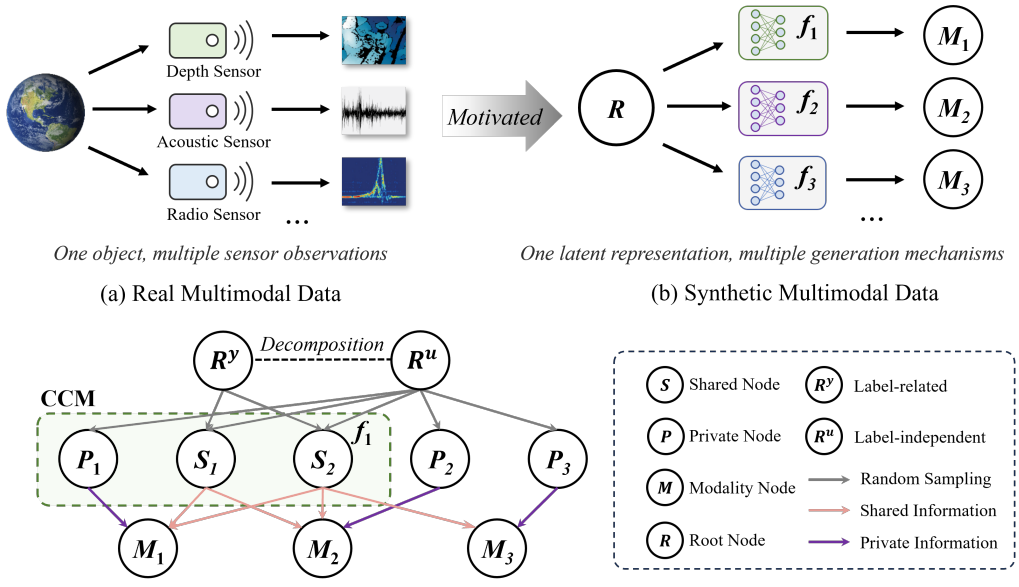


Figure 1: Synthetic multimodal dataset generation process. (a) Real-world multimodal observations are produced by different sensors perceiving the same underlying object or state. (b) Motivated by this process, synthetic modalities are generated from a common latent representation through distinct generation mechanisms in this paper. (c) An example SMCM with three modalities M_1, M_2, and M_3, where the root representation is decomposed into label-related and label-independent components, and the CCM organizes their allocation across shared and private nodes. Shared nodes connect to multiple modality nodes to introduce common information to different modalities, whereas each private node is associated exclusively with its corresponding modality to preserve modality-specific variation. Through different configurations of shared and private nodes, the SMCM induces diverse correlations among M_1, M_2, and M_3.

S_j and private nodes P_m to regulate information propagation across modalities. Specifically, shared nodes transmit selected information to multiple modality nodes, thereby inducing shared information across modalities, whereas each private node contributes modality-specific information only to its associated modality. By controlling the shared and private information pathways between the root and modality nodes, different SMCMs can induce diverse multimodal correlation patterns, enabling the systematic construction of diverse synthetic datasets.

Generation Mechanisms in SMCM. An SMCM consists of four types of nodes: a root node R, shared nodes S_j, private nodes P_m, and modality nodes M_m. Together, these nodes define a hierarchical generation process in which the root node provides the underlying latent information, the shared and private nodes regulate how different components of this information contribute to multiple or individual modalities, and the modality nodes represent the corresponding synthetic modality representations. To formally characterize the variable dependencies and generation processes involved in this construction, we adopt the formulation of Structural Causal Models (SCMs) [38], which is used to specify the generation of representations at the root and modality nodes.

An SCM provides a formal framework for describing causal dependencies and data-generating processes among variables. It specifies a directed acyclic graph (DAG) together with a collection of structural mechanisms [9, 10]. Root variables are independently sampled from their assigned distributions, while each non-root variable v_a is generated from its parent variables according to

$$v_a = f_a\left(\{g_{a,b}(v_b)\}_{v_b \in \mathrm{Pa}(v_a)}, \epsilon_a\right), \tag{1}$$

where $\mathrm{Pa}(v_a)$ denotes the set of parent variables of v_a, $g_{a,b}(\cdot)$ represents the edge function governing the transformation from parent variable v_b to v_a, $f_a(\cdot)$ denotes an aggregation function that combines

the transformed parent variables, and ϵ_a is an observational noise term. The diversity of variable dependencies is further controlled through the functional forms of the edge and aggregation functions. The edge functions can be instantiated using neural networks with different activation functions (e.g., sigmoid, ReLU, sine, and tanh), convolutional mappings, or decision-tree functions, while the aggregation function can adopt mean, weighted, or neural aggregation.

By varying these functional mechanisms, the SCM can represent a broad range of nonlinear dependencies and interaction patterns, providing flexible generation processes for the corresponding nodes in the SMCM. We next describe the different node types in the SMCM and their roles in constructing multimodal correlations.

Root Node. The root node R represents a high-dimensional latent representation shared across all modalities [39], from which shared nodes, private nodes, and modality nodes are subsequently generated. We partition the root representation into label-related and label-independent components to explicitly control task-related information while preserving other latent variations [40]. Specifically, for sample i, the root representation is defined as

$$\mathbf{r}_i = (\mathbf{h}_i + \alpha \boldsymbol{\mu}_{y_i})\mathbf{A} = [\mathbf{r}_i^y \| \mathbf{r}_i^u], \tag{2}$$

where $\mathbf{r}_i$ denotes the realization of root node R, $\mathbf{h}_i$ is the initial representation generated through the SCM process, $\boldsymbol{\mu}_{y_i}$ denotes the label-specific shift associated with class y_i, and α controls the strength of the label-related variation. The transformation matrix $\mathbf{A}$ aligns the label-related variation with a dedicated subspace, yielding the label-related component $\mathbf{r}_i^y$, while the remaining dimensions form the label-independent component $\mathbf{r}_i^u$. These two components are concatenated along the representation dimension, denoted by $\|$, to form the complete root representation $\mathbf{r}_i$.

Shared and Private Nodes. Shared and private nodes characterize common and modality-specific information in multimodal data, respectively. As illustrated in Figure 1(c), shared nodes S_j are constructed from the decomposed root representation and connect to multiple modality nodes, allowing shared information to influence different modalities. Specifically, each shared node S_j randomly selects a subset of dimensions from both the label-related component $\mathbf{r}_i^y$ and the label-independent component $\mathbf{r}_i^u$, denoted as Ω_y^j and Ω_u^j, respectively. The selected representations are transformed through randomly sampled sparse nonlinear mappings to obtain the shared node $\mathbf{s}_i^j$.

In contrast, private nodes P_m capture modality-specific information. For each modality m, the corresponding private node selects dimensions from the label-independent component $\mathbf{r}_i^u$ that are not assigned to shared nodes, denoted as Ω_p^m, and generates the modality-specific representation $\mathbf{p}_i^m$. Each private node is connected exclusively to its corresponding modality node, ensuring that the generated information only affects the associated modality.

Modality Nodes. Modality nodes M_m represent the modality-specific representations in each synthetic multimodal dataset. For each modality m, the corresponding modality node receives information from the root representation, the shared nodes connected to it, and its associated private node. Subsets of dimensions are randomly selected from these representations and combined as the input to a modality-specific generation mechanism. Accordingly, the representation of modality m for sample i is defined as follows.

$$\mathbf{x}_i^m = \mathcal{F}_m\left(\mathbf{r}_i[\Omega_r^m] \| \mathbf{s}_i[\Omega_s^m] \| \mathbf{p}_i^m[\Omega_p^m], \boldsymbol{\epsilon}_i^m\right), \tag{3}$$

where $\mathbf{s}_i$ denotes the combined representation of the shared nodes connected to modality node M_m, and Ω_r^m, Ω_s^m, and Ω_p^m denote the selected dimensions from the root, shared, and private representations, respectively. $\boldsymbol{\epsilon}_i^m$ denotes the random factors introduced during modality generation, and $\mathcal{F}_m(\cdot)$ denotes the SCM-based structural mechanism associated with modality node M_m. By sampling a large collection of SMCMs with diverse configurations, we construct a broad distribution of synthetic multimodal datasets for training the generalized multimodal foundation model.

3.3 Generalized Multimodal Foundation Model

As illustrated in Figure 2, our goal is to develop a unified model that can activate and utilize task-relevant multimodal correlation patterns from contextual examples, enabling direct prediction across diverse modality combinations and tasks. To this end, we train the model on a large collection of synthetic multimodal datasets generated by SMCMs, such that the model captures transferable multimodal correlation patterns rather than dataset-specific fusion rules.

We formulate each synthetic multimodal dataset as a context-conditioned prediction task. For a task with available modality subset $\mathcal{S} \subseteq \mathcal{M}$, the multimodal input of a sample is represented as $\mathbf{X} = (\mathbf{x}^m)_{m \in \mathcal{S}}$. Each task consists of a labeled context set $\mathcal{C} = \{(\mathbf{X}_i, y_i)\}_{i=1}^{K}$ and a set of query inputs $\{\mathbf{X}_j\}_{j=K+1}^{N}$. The context set provides task-specific information for identifying relevant multimodal correlations, while the query labels $\{y_j\}_{j=K+1}^{N}$ are not provided to the model and are used only as supervision during training. In this way, prediction is conditioned on the contextual examples rather than task-specific parameter adaptation, allowing the same model to generalize across heterogeneous multimodal tasks. The proposed model comprises unified modality token generation, hierarchical multimodal attention, and a multimodal in-context learning objective, which together support generalized multimodal prediction across diverse tasks.

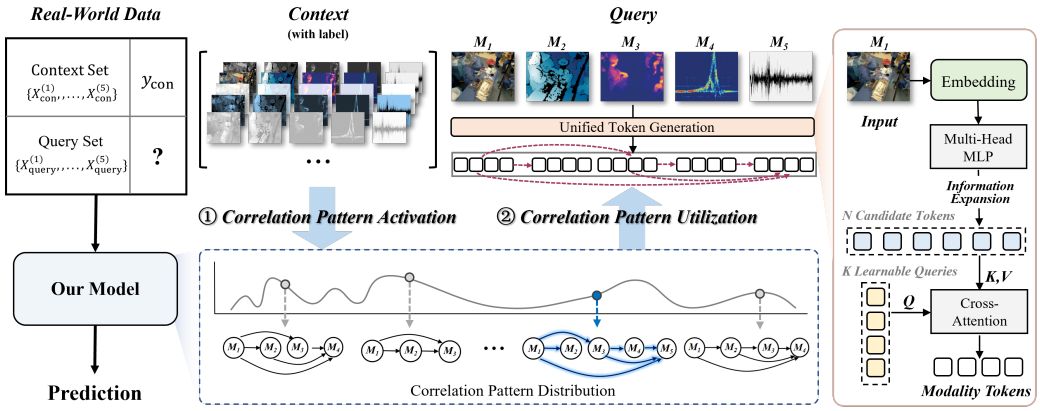


Figure 2: Overview of the proposed generalized multimodal foundation model. For the real-world multimodal data, observations from the available modalities are first encoded and mapped into a unified multimodal token space. The model, which internalizes a broad distribution of transferable multimodal correlation patterns during training, leverages labeled contextual examples to identify and activate the correlation pattern relevant to the current prefiction task. The activated pattern is then utilized to predict the query samples, while all model parameters remain fixed without task-specific adaptation.

Unified Token Generation. A key challenge in generalized multimodal learning is to construct a unified representation framework that can accommodate heterogeneous modality representations without relying on predefined modality combinations. To address this challenge, we introduce a unified token generation mechanism that maps modality-specific representations into a shared modality token space, providing a consistent representation form for subsequent multimodal modeling. Given the representation $\mathbf{x}_i^m$ associated with modality node M_m for sample i, the token generation process is formulated as

$$\mathcal{T}_i^m = \mathcal{G}_\phi(\mathbf{x}_i^m), \tag{4}$$

where $\mathcal{G}_\phi(\cdot)$ denotes the unified token generation mechanism parameterized by ϕ, and $\mathcal{T}_i^m = \{\mathbf{t}_{i,\ell}^m\}_{\ell=1}^{L_m}$ denotes the resulting token sequence for modality m. Each token $\mathbf{t}_{i,\ell}^m \in \mathbb{R}^d$ is represented in a common d-dimensional space, ensuring a consistent token representation across heterogeneous modalities.

The unified token generation mechanism follows a two-stage transformation process consisting of gated multi-head transformation and cross-attention aggregation. First, multiple transformation heads apply gated nonlinear mappings based on the Gated Linear Unit (GLU) to produce candidate representations from each modality. These candidates are then selectively aggregated through cross-attention with learnable query embeddings, yielding a compact sequence of modality tokens. The resulting tokens from all available modalities are combined into a unified token sequence for subsequent multimodal modeling.

Hierarchical Multimodal Attention. After converting modality-specific representations into unified modality tokens, we model the modality-level, cross-modality, and sample-level correlations based on the hierarchical multimodal attention mechanism.

- **Modality-Level Attention** captures dependencies among tokens within each modality, enabling the model to learn modality-specific representations within individual modality.

- **Cross-Modality Attention** models interactions among different modalities within each sample, allowing the model to capture complementary and shared predictive information across modalities.

- **Sample-Level Attention** aggregates information from contextual samples, enabling the model to infer task-specific predictive patterns and adapt to unseen multimodal tasks through in-context learning.

Through hierarchical attention, the model preserves the structured relationships among modalities and samples instead of treating multimodal tokens as an unordered sequence, facilitating the learning of transferable multimodal correlations across diverse tasks.

Multimodal In-context Learning Objective. The proposed generalized multimodal foundation model is trained under the in-context learning paradigm, where each synthetic multimodal dataset $\mathcal{D}$ is treated as an independent prediction task. Unlike conventional fusion methods that optimize a dedicated model for each dataset, our model is trained across diverse synthetic multimodal tasks with different modality combinations and causal structures. For each task, the labeled context set $\mathcal{C}$ provides task-specific information for identifying relevant multimodal correlations, while the query set $\mathcal{Q}$ provides the prediction targets used for optimization. The training objective is defined over synthetic multimodal datasets sampled from the generation distribution:

$$\mathcal{L}(\theta) = -\mathbb{E}_{\mathcal{D}\sim p_{\mathrm{syn}}(\mathcal{D})}\left[\frac{1}{|\mathcal{Q}|}\sum_{(\mathbf{X}_i,y_i)\in\mathcal{Q}}\log p_\theta(y_i \mid \mathbf{X}_i,\mathcal{C})\right], \tag{5}$$

where $p_{\mathrm{syn}}(\mathcal{D})$ denotes the distribution over synthetic multimodal datasets induced by the proposed generation process, and θ denotes the trainable parameters of the model. By optimizing this objective across diverse synthetic multimodal tasks, the model learns transferable multimodal correlation patterns rather than fitting a task-specific fusion function to each dataset.

Parameter-efficient Multimodal Extension. Our model inherits the context-conditioned inference paradigm of TabPFN [9] and extends it from structured data to generalized multimodal tasks. Rather than modifying the entire pretrained model, we introduce Low-Rank Adaptation (LoRA) [41] into the attention layers and jointly optimize the LoRA parameters, unified token generation module, and prediction head, while keeping the remaining pretrained parameters frozen. This parameter-efficient extension enables the model to acquire multimodal correlation modeling capability from large-scale synthetic multimodal datasets while preserving the context-based inference mechanism for prediction on unseen tasks.

3.4 Inference without Adaptation

After training on diverse synthetic multimodal tasks, the proposed generalized multimodal foundation model can be directly applied to real-world datasets. Exposure to diverse datasets enables the model to internalize a broad space of multimodal correlation patterns, from which task-relevant patterns can be identified and activated through contextual examples during inference.

For a real-world dataset, observations from each available modality are first encoded by suitable pre-trained modality encoders and subsequently mapped into the unified token space. Since the model concentrates on the modality representations rather than encoder-specific architectures, it remains flexible with respect to the choice of pre-trained encoders. Given a labeled real-world context set $\mathcal{C}^{\mathrm{real}}$ and an unlabeled query instance $\mathbf{X}_q^{\mathrm{real}}$, the model predicts its target according to $p_\theta(y_q^{\mathrm{real}} \mid \mathbf{X}_q^{\mathrm{real}}, \mathcal{C}^{\mathrm{real}})$ while keeping all model parameters fixed. This enables direct prediction on unseen multimodal datasets with diverse modality combinations and prediction tasks without task-specific optimization.

4 Experiments

4.1 Experimental Settings

Datasets and Evaluation Metrics. We evaluate our model on a diverse collection of real-world multimodal datasets covering various application domains, such as social media analysis, affective computing, healthcare, and biomedical analysis. Detailed dataset descriptions, including modality configurations and data splits, are provided in the Appendix. We report Accuracy (ACC) and Area Under the Receiver Operating Characteristic Curve (AUC) as evaluation metrics. For supervised multimodal fusion baselines, models are trained on the training set, tuned on the validation set, and evaluated on the test set. In contrast, our model follows a multimodal in-context learning paradigm, where training and validation samples are provided as contextual examples and test samples are used as query instances without additional parameter optimization or task-specific adaptation.

Comparison Methods. We compare our model with representative supervised multimodal fusion methods from major fusion paradigms, including feature-level fusion, interaction-based fusion, hierarchical fusion, and decision-level fusion. Concat serves as a simple feature-level fusion baseline that directly concatenates modality representations before prediction [17]. EFT represents Transformer-based fusion methods, which employ a unified encoder to model interactions among multimodal representations through self-attention mechanisms [18]. CAF is an interaction-based fusion approach that leverages cross-attention to explicitly capture information exchange between different modalities [29]. Multi-to-One adopts a hierarchical fusion strategy, where modality-specific encoders first extract individual representations and a shared fusion encoder subsequently integrates multimodal information [42]. TMC represents decision-level fusion methods by combining modality-specific predictions through evidential uncertainty modeling [24]. These methods provide strong supervised baselines for evaluating whether our generalized model can achieve comparable performance to task-specific multimodal fusion approaches without additional adaptation.

Model Training. We train our model following the in-context learning paradigm, where each synthetic multimodal dataset is treated as an independent prediction task. During training, we perform approximately $20,000$ optimization steps, with each step sampling 16 synthetic multimodal datasets, resulting in around $320,000$ unique synthetic tasks. The number of samples in each dataset is randomly generated within the range of 512 to $2,048$, and each task is further divided into a context set and a query set, where the query size is fixed to 128 while the remaining samples serve as contextual examples. We optimize the model using the Adam optimizer with a learning rate of 2×10^{-4}, employing a linear warm-up phase followed by cosine annealing for learning-rate scheduling.

4.2 Main Results

Figure 3 summarizes the performance ranking of our model and representative specialized fusion methods across real-world multimodal tasks. Existing fusion methods exhibit substantial performance variations across different datasets, suggesting that their effectiveness is highly dependent on specific modality combinations and task characteristics. For relatively simple tasks or scenarios where modalities contain highly redundant information, simple feature-level fusion methods such as Concat can be sufficient, leaving limited gains for more sophisticated fusion architectures. In contrast, when complementary information across modalities needs to be effectively captured, interaction-based methods such as CAF can achieve stronger performance by modeling cross-modal correlations. However, **no specialized fusion method performs consistently well across all modality combinations and prediction tasks**, indicating their limited ability to generalize beyond predefined settings.

We further present the detailed comparison results across different datasets. As shown in Figure 4, **The proposed generalized multimodal foundation model achieves performance comparable to state-of-the-art specialized methods without parameter adaptation**, improving the average ACC by 5.83% and AUC by 3.10% over existing specialized fusion methods. Notably, these results are obtained without any task-specific optimization or parameter updates during inference, demonstrating the capability of our model to generalize beyond individual multimodal tasks. Unlike specialized fusion models that optimize dedicated fusion strategies for predefined modality combinations and datasets, our model is trained on large-scale synthetic multimodal datasets with diverse causal structures, enabling it to capture the underlying distribution of multimodal correlation patterns rather than specific fusion functions. During inference, contextual examples from real datasets provide

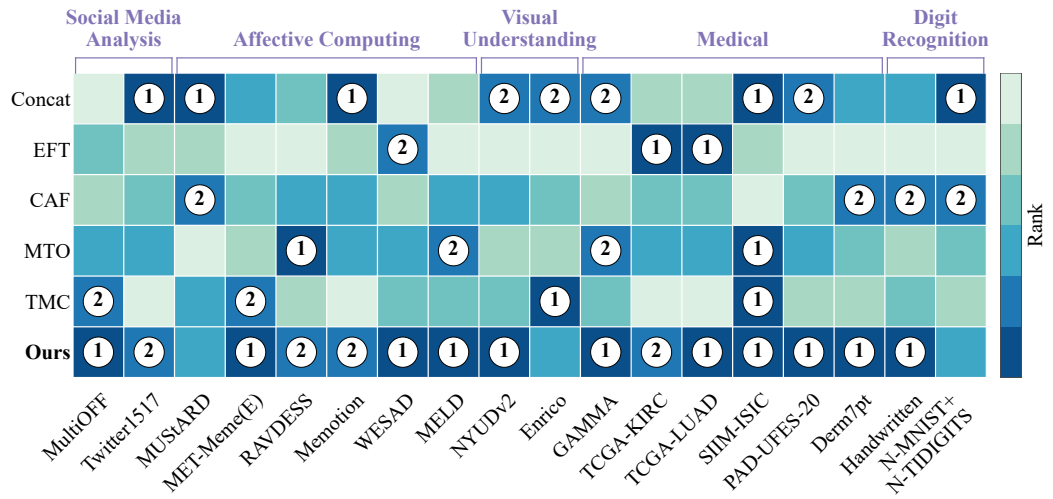


Figure 3: Comparison of performance ranking between our generalized multimodal foundation model and representative specialized fusion methods across diverse multimodal datasets. Our model achieves performance comparable to state-of-the-art specialized methods without parameter adaptation.

task-specific information, allowing the model to activate the relevant correlation pattern and utilize it for effective prediction across diverse modality combinations and downstream tasks.

4.3 Ablation Study

Effectiveness of Multimodal Extension. To evaluate the effectiveness of the proposed model in extending context-based prediction toward generalized multimodal scenarios, we compare it with TabPFN under the same input settings. Here, we first apply PCA to the raw modality representations and project them into the same token space, ensuring that both models receive the same number of input tokens for a fair comparison. As shown in Table 1, our model exhibits improved overall performance across diverse datasets, indicating its enhanced capability in modeling multimodal correlations. These results demonstrate that training over diverse synthetic multimodal datasets enables the model to acquire richer multimodal correlation patterns and effectively extend the prediction capability of the base model to heterogeneous multimodal scenarios. For more comparison results, refer to Appendix.

Table 1: Comparison of the original TabPFN and the generalized multimodal foundation model.

Method	GAMMA		TCGA-KIRC		Enrico		MET-Meme(E)		RAVDESS		WESAD		TCGA-LUAD		PAD-UFES-20	
	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑
TabPFN	65.00	80.67	69.70	69.83	49.66	80.75	35.55	64.16	55.21	88.25	79.34	90.90	54.10	**58.11**	76.29	**93.16**
Ours	**70.00**	**83.33**	**72.73**	**71.49**	**50.68**	**83.51**	**38.39**	**66.02**	**60.00**	**92.51**	**83.35**	**93.35**	**60.66**	45.16	**76.85**	93.04

Flexibility in the Choice of Pre-trained Encoders. To demonstrate that the generalized multimodal foundation model is not restricted to particular modality encoders, we evaluate its performance under diverse pre-trained encoder configurations. As reported in Table 2, our model maintains strong prediction performance across different configurations, outperforming the average performance of specialized fusion methods by 9.61 percentage points across all evaluated encoder settings. Notably, the advantage is preserved across different modality and encoder combinations, suggesting that the learned multimodal correlations can be effectively utilized despite variations in how individual modalities are encoded. Although the quality of extracted representations inevitably affects the final performance, our model is able to activate and utilize appropriate multimodal correlation patterns learned during training, achieving comparable or superior results to specialized fusion methods in most cases. These results indicate that our model is not coupled with particular encoders, but instead learns transferable multimodal correlation patterns from synthetic datasets.

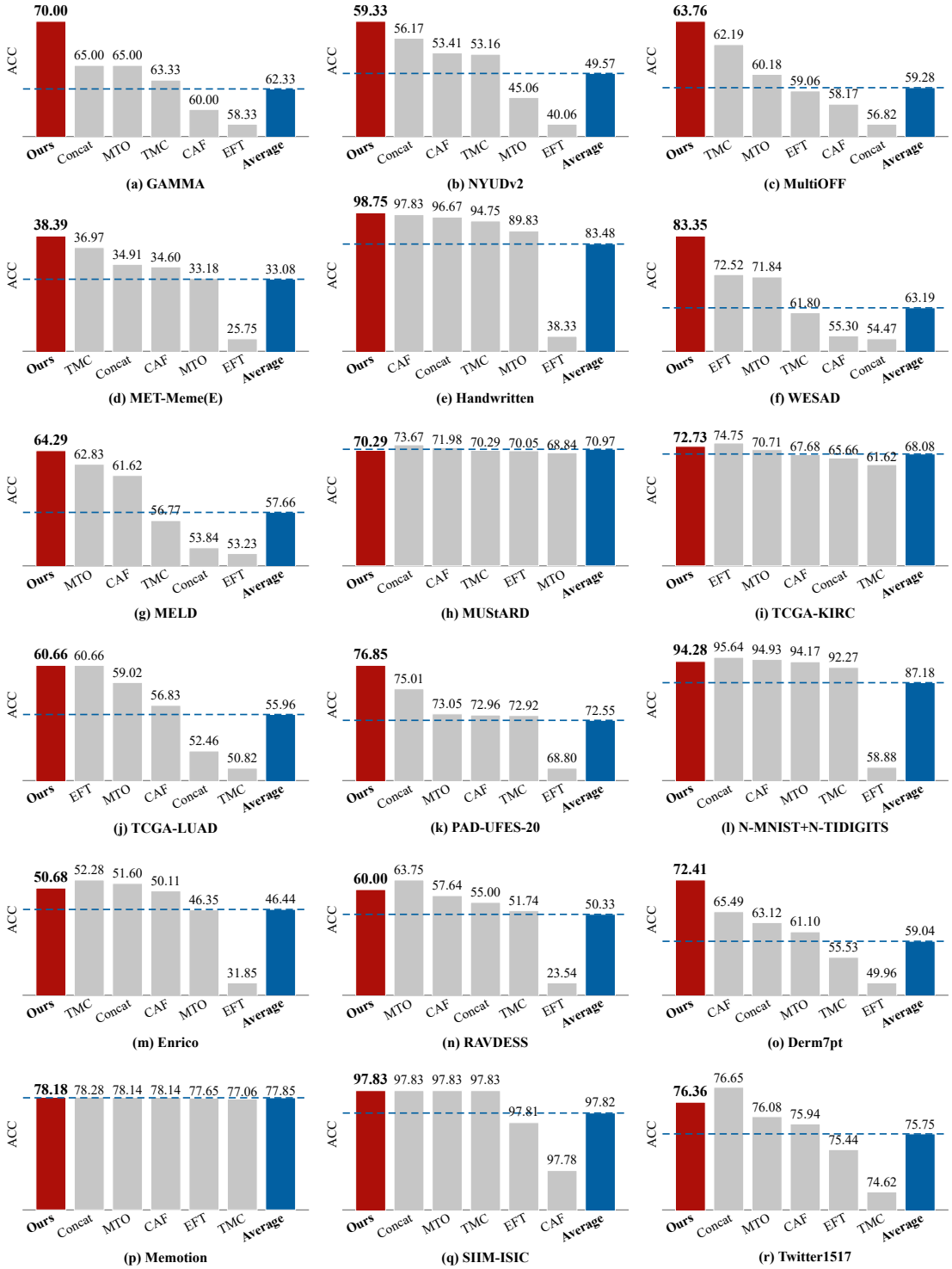


Figure 4: Comparison between our generalized multimodal foundation model and specialized fusion models across diverse multimodal datasets.

Table 2: Comparison under different pre-trained encoder configurations. For each encoder configuration, the best and second-best results among all methods are highlighted in bold and underlined, respectively.

Dataset	Modalities	Combination	Concat	EFT	MTO	CAF	TMC	Avg.	Ours
Derm7pt	Image+Tabular	ResNet18+TabPFN	61.10	53.92	62.87	63.80	56.29	59.60	**72.41**
		DINOv3+TabPFN	66.50	54.43	58.73	64.56	60.42	60.93	**74.68**
MultiOFF	Image+Text	DINOv3+Qwen3	56.82	59.06	60.18	58.17	62.19	59.28	**63.76**
		ResNet18+Qwen3	56.38	61.74	**64.21**	56.15	62.64	60.22	63.76
		DINOv3+SigLIP2	55.03	58.39	61.07	58.61	54.59	57.54	**61.74**
NYUDv2	RGB+Depth	ResNet18+ResNet18	56.17	40.06	45.06	53.41	53.16	49.57	**59.33**
		DINOv3+ResNet18	73.04	41.79	64.93	70.74	71.61	64.42	**75.54**
		ResNet18+DINOv3	60.60	31.09	52.04	58.51	58.10	52.07	**62.23**
RAVDESS	Audio+Visual	WavLM+DINOv3	55.00	23.54	**63.75**	57.64	51.74	50.33	60.00
		Whisper+DINOv3	**76.53**	35.42	66.04	67.22	67.71	62.58	74.17
		WavLM+ResNet18	31.04	19.17	**51.39**	35.00	32.99	33.92	48.54

Effect of Contextual Examples.

To investigate the impact of contextual examples on prediction performance, we evaluate our model under different context fractions while keeping the query set fixed. Specifically, contextual examples are randomly sampled from the available context pool with three random seeds $\{1, 2, 3\}$ at fractions ranging from 10% to 100%. As shown in Figure 5, the model exhibits an overall positive scaling trend across diverse multimodal datasets. The prediction performance generally improves as more contextual examples become available, while the marginal gains tend to diminish at larger context fractions. This trend indicates that the model can effectively leverage contextual examples to identify task-relevant multimodal correlation patterns, while maintaining effective prediction even with limited contextual information. These results further demonstrate the stability of the model's in-context prediction capability across varying context sizes.

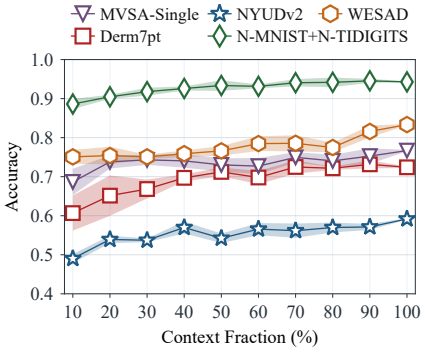


Figure 5: Performance scaling with different fractions of contextual examples.

5 Conclusion

In this paper, we introduce the Generalized Multimodal Foundation Model, a new paradigm toward building a unified multimodal prediction model beyond predefined modality combinations and task settings. By constructing large-scale synthetic datasets with diverse causal structures, we enable the model to capture general multimodal correlation patterns rather than relying on task-specific fusion strategies. Through multimodal in-context learning, the proposed model can leverage contextual examples to activate relevant correlation patterns and perform prediction without additional parameter updates. Extensive experiments demonstrate that our model achieves competitive performance with state-of-the-art specialized fusion models across various multimodal benchmarks, despite requiring no task-specific optimization during inference. This work provides a promising direction toward supporting future modalities beyond those explored by existing multimodal learning paradigms.

References

[1] Tadas Baltrušaitis, Chaitanya Ahuja, and Louis-Philippe Morency. Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2):423–443, 2018.

[2] Peng Xu, Xiatian Zhu, and David A Clifton. Multimodal learning with transformers: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10):12113–12132, 2023.

[3] Qingyang Zhang, Yake Wei, Zongbo Han, Huazhu Fu, Xi Peng, Qinghua Hu, Cheng Deng, Cai Xu, Jie Wen, Di Hu, et al. Multimodal fusion on low-quality data: A comprehensive survey. *Information Fusion*, page 104437, 2026.

[4] Paul Pu Liang, Yun Cheng, Xiang Fan, Chun Kai Ling, Suzanne Nie, Richard Chen, Zihao Deng, Nicholas Allen, Randy Auerbach, Faisal Mahmood, et al. Quantifying & modeling multimodal interactions: An information decomposition framework. *Advances in Neural Information Processing Systems*, 36:27351–27393, 2023.

[5] Albert Haque, Arnold Milstein, and Li Fei-Fei. Illuminating the dark spaces of healthcare with ambient intelligence. *Nature*, 585(7824):193–202, 2020.

[6] Qingyang Zhang, Haitao Wu, Changqing Zhang, Qinghua Hu, Huazhu Fu, Joey Tianyi Zhou, and Xi Peng. Provable dynamic fusion for low-quality multimodal data. In *International Conference on Machine Learning*, pages 41753–41769, 2023.

[7] Xi Wei, Tianzhu Zhang, Yan Li, Yongdong Zhang, and Feng Wu. Multi-modality cross attention network for image and sentence matching. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10938–10947, 2020.

[8] Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.

[9] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025.

[10] Yuanrui Wang, Xingxuan Zhang, Han Yu, Mingchao Hao, Gang Ren, Hao Yuan, Li Mao, Yunjia Zhang, Chun Yuan, and Peng Cui. Limix-2m: Mitigating low-rank collapse and attention bottlenecks in tabular foundation models. *arXiv preprint arXiv:2606.04485*, 2026.

[11] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.

[12] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.

[13] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C Berg, Wan-Yen Lo, et al. Segment anything. In *2023 IEEE/CVF international conference on computer vision (ICCV)*, pages 3992–4003. IEEE, 2023.

[14] Paul Pu Liang, Amir Zadeh, and Louis-Philippe Morency. Foundations & trends in multimodal machine learning: Principles, challenges, and open questions. *ACM Computing Surveys*, 56(10):1–42, 2024.

[15] Xinlong Wang, Yufeng Cui, Jinsheng Wang, Fan Zhang, Yueze Wang, Xiaosong Zhang, Zhengxiong Luo, Quan Sun, Zhen Li, Yuqi Wang, et al. Multimodal learning with next-token prediction for large multimodal models. *Nature*, 650(8101):327–333, 2026.

[16] Hao Dong, Moru Liu, Kaiyang Zhou, Eleni Chatzi, Juho Kannala, Cyrill Stachniss, and Olga Fink. Advances in multimodal adaptation and generalization: From traditional approaches to foundation models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026.

[17] Leyan Xue, Changqing Zhang, Kecheng Xue, Xiaohong Liu, Guangyu Wang, and Zongbo Han. MULTIBENCH++: A unified and comprehensive multimodal fusion benchmarking across specialized domains. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 27450–27458, 2026.

[18] Paul Pu Liang, Yiwei Lyu, Xiang Fan, Zetian Wu, Yun Cheng, Jason Wu, Leslie Chen, Peter Wu, Michelle A Lee, Yuke Zhu, et al. Multibench: Multiscale benchmarks for multimodal representation learning. *Advances in Neural Information Processing Systems*, 2021(DB1):1, 2021.

[19] Zongbo Han, Fan Yang, Junzhou Huang, Changqing Zhang, and Jianhua Yao. Multimodal dynamics: Dynamical fusion for trustworthy multimodal classification. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20675–20685. IEEE, 2022.

[20] Changqing Zhang, Zongbo Han, Huazhu Fu, Joey Tianyi Zhou, Qinghua Hu, et al. CPM-Nets: Cross partial multi-view networks. *Advances in neural information processing systems*, 32, 2019.

[21] Wall Kim, Chaeyoung Song, and Hanul Kim. Multimodalpfn: Extending prior-data fitted networks for multimodal tabular learning. *arXiv preprint arXiv:2602.20223*, 2026.

[22] Dhanesh Ramachandram and Graham W Taylor. Deep multimodal learning: A survey on recent advances and trends. *IEEE Signal Processing Magazine*, 34(6):96–108, 2017.

[23] Pradeep K Atrey, M Anwar Hossain, Abdulmotaleb El Saddik, and Mohan S Kankanhalli. Multimodal fusion for multimedia analysis: a survey. *Multimedia Systems*, 16(6):345–379, 2010.

[24] Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou. Trusted multi-view classification with dynamic evidential fusion. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2):2551–2566, 2022.

[25] Rohit Girdhar, Mannat Singh, Nikhila Ravi, Laurens Van Der Maaten, Armand Joulin, and Ishan Misra. Omnivore: A single model for many visual modalities. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16081–16091. IEEE, 2022.

[26] Yikai Wang, Xinghao Chen, Lele Cao, Wenbing Huang, Fuchun Sun, and Yunhe Wang. Multimodal token fusion for vision transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12176–12185. IEEE, 2022.

[27] Arsha Nagrani, Shan Yang, Anurag Arnab, Aren Jansen, Cordelia Schmid, and Chen Sun. Attention bottlenecks for multimodal fusion. *Advances in Neural Information Processing Systems*, 34:14200–14213, 2021.

[28] Licai Sun, Bin Liu, Jianhua Tao, and Zheng Lian. Multimodal cross-and self-attention network for speech emotion recognition. In *IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 4275–4279. IEEE, 2021.

[29] Xunlin Zhan, Yangxin Wu, Xiao Dong, Yunchao Wei, Minlong Lu, Yichi Zhang, Hang Xu, and Xiaodan Liang. Product1m: Towards weakly supervised instance-level product retrieval via cross-modal pretraining. In *IEEE/CVF International Conference on Computer Vision*, pages 11762–11771, 2021.

[30] Yiran Huang, Karsten Roth, Quentin Bouniot, Wenjia Xu, and Zeynep Akata. Dissecting multimodal in-context learning: Modality asymmetries and circuit dynamics in modern transformers. *arXiv preprint arXiv:2601.20796*, 2026.

[31] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763. PmLR, 2021.

[32] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision. In *International Conference on Machine Learning*, pages 4904–4916. PMLR, 2021.

[33] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katherine Millican, Malcolm Reynolds, et al. Flamingo: a visual language model for few-shot learning. *Advances in Neural Information Processing Systems*, 35:23716–23736, 2022.

[34] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on Machine Learning*, pages 19730–19742, 2023.

[35] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in Neural Information Processing Systems*, 36:34892–34916, 2023.

[36] Chameleon Team. Chameleon: Mixed-modal early-fusion foundation models, 2024. *URL https://arxiv. org/abs/2405.09818*, 9(8), 2024.

[37] Chengyue Wu, Xiaokang Chen, Zhiyu Wu, Yiyang Ma, Xingchao Liu, Zizheng Pan, Wen Liu, Zhenda Xie, Xingkai Yu, Chong Ruan, et al. Janus: Decoupling visual encoding for unified multimodal understanding and generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12966–12977. IEEE, 2025.

[38] Judea Pearl. *Causality*. Cambridge university press, 2009.

[39] Changqing Zhang, Huazhu Fu, Qinghua Hu, Xiaochun Cao, Yuan Xie, Dacheng Tao, and Dong Xu. Generalized latent multi-view subspace clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(1):86–99, 2018.

[40] Brooks Paige, Jan-Willem Van De Meent, Alban Desmaison, Noah Goodman, Pushmeet Kohli, Frank Wood, Philip Torr, et al. Learning disentangled representations with semi-supervised deep generative models. *Advances in Neural Information Processing Systems*, 30, 2017.

[41] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.

[42] Ruilong Li, Shan Yang, David A Ross, and Angjoo Kanazawa. Ai choreographer: Music conditioned 3d dance generation with aist++. In *IEEE/CVF International Conference on Computer Vision*, pages 13381–13392, 2021.

[43] Junde Wu, Huihui Fang, Fei Li, Huazhu Fu, Fengbin Lin, Jiongcheng Li, Yue Huang, Qinji Yu, Sifan Song, Xinxing Xu, et al. Gamma challenge: glaucoma grading from multi-modality images. *Medical Image Analysis*, 90:102938, 2023.

[44] Veronica Rotemberg, Nicholas Kurtansky, Brigid Betz-Stablein, Liam Caffery, Emmanouil Chousakos, Noel Codella, Marc Combalia, Stephen Dusza, Pascale Guitera, David Gutman, et al. A patient-centric dataset of images and metadata for identifying melanomas using clinical context. *Scientific Data*, 8(1):34, 2021.

[45] Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. Indoor segmentation and support inference from rgbd images. In *European Conference on Computer Vision*, pages 746–760. Springer, 2012.

[46] Garrick Orchard, Ajinkya Jayawant, Gregory K Cohen, and Nitish Thakor. Converting static image datasets to spiking neuromorphic datasets using saccades. *Frontiers in Neuroscience*, 9:437, 2015.

[47] Jithendar Anumula, Daniel Neil, Tobi Delbruck, and Shih-Chii Liu. Feature representations for neuromorphic audio spike streams. *Frontiers in Neuroscience*, 12:23, 2018.

[48] John N Weinstein, Eric A Collisson, Gordon B Mills, Kenna R Shaw, Brad A Ozenberger, Kyle Ellrott, Ilya Shmulevich, Chris Sander, and Joshua M Stuart. The cancer genome atlas pan-cancer analysis project. *Nature Genetics*, 45(10):1113–1120, 2013.

[49] Santiago Castro, Devamanyu Hazarika, Verónica Pérez-Rosas, Roger Zimmermann, Rada Mihalcea, and Soujanya Poria. Towards multimodal sarcasm detection (an obviously perfect paper). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4619–4629, 2019.

[50] Luis A Leiva, Asutosh Hota, and Antti Oulasvirta. Enrico: A dataset for topic modeling of mobile ui designs. In *International Conference on Human-Computer Interaction with Mobile Devices and Services*, pages 1–4, 2020.

[51] Shardul Suryawanshi, Bharathi Raja Chakravarthi, Mihael Arcan, and Paul Buitelaar. Multimodal meme dataset (multioff) for identifying offensive content in image and text. In *Proceedings of the second workshop on trolling, aggression and cyberbullying*, pages 32–41, 2020.

[52] Bo Xu, Tingting Li, Junzhe Zheng, Mehdi Naseriparsa, Zhehuan Zhao, Hongfei Lin, and Feng Xia. Met-meme: A multimodal meme dataset rich in metaphors. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2887–2899, 2022.

[53] Steven R Livingstone and Frank A Russo. The ryerson audio-visual database of emotional speech and song (ravdess): A dynamic, multimodal set of facial and vocal expressions in north american english. *PloS one*, 13(5):e0196391, 2018.

[54] Chhavi Sharma, Deepesh Bhageria, William Scott, Srinivas Pykl, Amitava Das, Tanmoy Chakraborty, Viswanath Pulabaigari, and Björn Gambäck. Semeval-2020 task 8: Memotion analysis-the visuo-lingual metaphor! In *Proceedings of the Workshop on Semantic Evaluation*, pages 759–773, 2020.

[55] Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger, and Kristof Van Laerhoven. Introducing wesad, a multimodal dataset for wearable stress and affect detection. In *Proceedings of the 20th ACM International Conference on Multimodal Interaction*, pages 400–408, 2018.

[56] Elisabetta Fersini, Francesca Gasparini, Giulia Rizzi, Aurora Saibene, Berta Chulvi, Paolo Rosso, Alyssa Lees, and Jeffrey Sorensen. Semeval-2022 task 5: Multimedia automatic misogyny identification. In *Proceedings of the 16th International Workshop on Semantic Evaluation*, pages 533–549, 2022.

[57] Andre GC Pacheco, Gustavo R Lima, Amanda S Salomao, Breno Krohling, Igor P Biral, Gabriel G De Angelo, Fábio CR Alves Jr, José GM Esgario, Alana C Simora, Pedro BC Castro, et al. Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32:106221, 2020.

A Dataset Details

GAMMA [43]

Data Source and Content From the Glaucoma grAding from Multi-Modality imAges challenge. It contains color fundus images and OCT volumes for glaucoma diagnosis.

Link https://zenodo.org/records/15119049

Feature Extraction Both the fundus images and OCT volumes are encoded using a CNNEncoder.

Data Splits A stratified split with seed 42 first assigns 80 samples to the non-test partition and 20 samples to the test set. The non-test partition is then divided using another stratified split with seed 10, yielding 64 training, 16 validation, and 20 test samples.

SIIM-ISIC [44]

Data Source and Content From the 2020 SIIM-ISIC Melanoma Classification Kaggle challenge. The dataset contains thousands of dermoscopic images of skin lesions, with patient-level metadata, for classifying lesions.

Link https://www.kaggle.com/competitions/siim-isic-melanoma-classification/data

Feature Extraction A ResNet encoder is used for the dermoscopic images, and an independent identity mapping encoder is used for the tabular patient metadata.

Data Splits We adopt an 8:1:1 split for training, validation, and test sets, containing 26,502 training, 3,312 validation, and 3,312 test utterances, with the random seed fixed at 42.

NYUDv2 [45]

Data Source and Content The NYU-Depth Dataset V2 provides RGB and depth images of various indoor scenes captured using a Microsoft Kinect. We use 10 scene categories for classification.

Link https://cs.nyu.edu/~fergus/datasets/nyu_depth_v2.html

Feature Extraction A ResNet encoder is used for the RGB images and another for the depth images.

Data Splits The dataset is split into 381 training, 414 validation, and 654 test samples.

N-MNIST+N-TIDIGITS [46, 47]

Data Source and Content Neuromorphic versions of MNIST (vision) and TIDIGITS (audio-spoken digits). Data are recorded as asynchronous event streams from event-based sensors. Digits 1–9 are used for 9-class classification.

Feature Extraction We follow https://github.com/MrLinNing/MemristorLSM to form aligned image–sound pairs. Deterministic spatial and temporal event descriptors are extracted from both modalities.

Data Splits Each class is randomly divided into 70% training, 15% validation, and 15% test sets with seed 10, containing 2,835 training, 603 validation, and 612 test utterances.

TCGA-KIRC [48]

Data Source and Content TCGA-KIRC contains aligned molecular measurements for kidney renal clear cell carcinoma patients. Six omics views are used for binary survival-status classification.

Link https://portal.gdc.cancer.gov/projects/TCGA-KIRC

Feature Extraction CNV, DNA methylation, miRNA, RNA-seq, RPPA, and cell-composition measurements are used directly. Each omics modality is represented using an identity mapping encoder.

Data Splits A fixed non-stratified split with seed 10 is used, containing 92 training, 41 validation, and 33 test patients.

TCGA-LUAD [48]

Data Source and Content TCGA-LUAD contains aligned molecular measurements for lung adenocarcinoma patients. Six omics views are used for binary survival-status classification.

Link https://portal.gdc.cancer.gov/projects/TCGA-LUAD

Feature Extraction CNV, DNA methylation, miRNA, RNA-seq, RPPA, and cell-composition measurements are used directly. Each omics modality is represented using an identity mapping encoder.

Data Splits A fixed non-stratified split with seed 10 is used, containing 169 training, 76 validation, and 61 test patients.

MUStARD [49]

Data Source and Content MUStARD contains aligned text, audio, and video utterances for binary sarcasm detection.

Link https://github.com/soujanyaporia/MUStARD

Feature Extraction The officially released BERT, librosa, and ResNet-152 features are used for text, audio, and video, respectively. Temporal audio and video features are averaged within each utterance.

Data Splits Fold 1 of the official speaker-dependent stratified five-fold split is used. The remaining 552 utterances are divided with seed 10 into 441 training and 111 validation samples, while 138 utterances form the test set.

Enrico [50]

Data Source and Content Enrico contains mobile UI screenshots and corresponding semantic wireframes for interface-topic classification. The original 20 design topics are merged into ten functional classes.

Link https://github.com/luileito/enrico

Feature Extraction Screenshots and semantic wireframes are independently encoded using an ImageNet-pretrained ResNet-18.

Data Splits A fixed 80:20 split with seed 0 is used, with 15% of the training portion reserved for validation, containing 992 training, 176 validation, and 292 test interfaces.

MultiOFF [51]

Data Source and Content MultiOFF contains image–text posts for binary offensive-content detection.

Link https://github.com/bharathichezhiyan/Multimodal-Meme-Classification-Identifying-Offensive-Content-in-Image-and-Text

Feature Extraction Images are encoded using a frozen DINOv3 ViT-B/16, and text is encoded using Qwen3-Embedding-0.6B.

Data Splits The dataset is split into 445 training, 149 validation, and 149 test posts.

MET-Meme(E) [52]

Data Source and Content The English subset of MET-Meme contains image–text memes for seven-class sentiment classification.

Link https://github.com/liaolianfoka/MET-Meme-A-Multi-modal-Meme-Dataset-Rich-in-Metaphors

Feature Extraction A ResNet encoder is used for the images, and a BERT model is used for the text.

Data Splits A 7:1:2 split with seed 42 is used, containing 737 training, 105 validation, and 211 test samples.

Twitter2015 [17]

Data Source and Content Originating from SemEval tasks, Twitter2015 pairs tweets with relevant images for three-class multimodal aspect-based sentiment analysis.

Link https://archive.org/details/twitterstream

Feature Extraction A ResNet encoder is used for the images, and a BERT model is used for the text.

Data Splits The dataset is split into 3,179 training, 1,122 validation, and 1,037 test samples.

Twitter1517 [17]

Data Source and Content Twitter1517 contains paired tweet text and images for binary image–text correlation classification.

Link https://github.com/code-chendl/HFIR

Feature Extraction Images are encoded using a frozen DINOv3 ViT-B/16, and text is encoded using Qwen3-Embedding-0.6B.

Data Splits A stratified 7:1:2 split with seed 42 is used, containing 3,270 training, 467 validation, and 935 test samples.

Handwritten [24]

Data Source and Content Handwritten is derived from the UCI Multiple Features dataset, which contains $2,000$ handwritten-digit samples from ten classes (0–9), with 200 samples per class. Each sample is represented by six types of descriptors as different views.

Link https://github.com/Han-Zongbo/TMC/tree/main/TMC%20ICLR/datasets

Feature Extraction We use the six released descriptor views directly: 240-dimensional pixel features, 76-dimensional Fourier features, 216-dimensional profile features, 47-dimensional Zernike features, 64-dimensional Karhunen–Loève features, and six-dimensional morphological features.

Data Splits The official 1,600 training samples are stratified with seed 10 into 1,280 training and 320 validation samples, while the 400 official test samples remain unchanged. The training and validation sets are combined to form the Context set.

RAVDESS [53]

Data Source and Content The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) contains recordings from 24 professional actors. We use its speech subset and pair each official audio-only recording with the corresponding video-only recording for eight-class emotion recognition: neutral, calm, happy, sad, angry, fearful, disgust, and surprised.

Link https://zenodo.org/records/1188976

Feature Extraction Audio recordings are encoded using WavLM Base+, while video frames are encoded using DINOv3 ViT-B/16 and averaged to obtain visual representations.

Data Splits We use a fixed actor-disjoint split. Actors 4, 10, 16, and 22 form the 240 validation utterances; actors 5, 6, 11, 12, 17, 18, 23, and 24 form the 480 test utterances; and the remaining actors form the 720 training utterances. The training and validation sets are combined to form the Context set.

Memotion [54]

Data Source and Content Memotion contains memes with transcribed text and affective annotations. We use its binary humor task, treating not-funny as the negative class and merging funny, very funny, and hilarious into the positive class.

Link https://www.kaggle.com/datasets/williamscott701/memotion-dataset-7k

Feature Extraction Meme images are encoded using DINOv3 ViT-B/16, while transcribed text is encoded using Qwen3-Embedding-0.6B.

Data Splits After removing samples with missing text, $6,831$ samples remain. We adopt a fixed, non-stratified $8:1:1$ split for the training, validation, and test sets, containing $5,465$ training, 683 validation, and 683 test samples, with the random seed fixed at 42. The training and validation sets are combined to form the $6,148$ context samples.

WESAD [55]

Data Source and Content WESAD is a wearable stress-and-affect dataset recorded from 15 subjects using chest- and wrist-worn sensors. We retain the baseline (neutral), stress, and amusement states for three-class classification.

Link https://archive.ics.uci.edu/dataset/465/wesad

Feature Extraction Chest- and wrist-sensor signals are independently encoded using MOMENT-1-small.

Data Splits Each contiguous retained-state interval is divided into non-overlapping 60-second windows, yielding 535 windows. We use deterministic 15-fold leave-one-subject-out evaluation. In each fold, one subject forms the Test set and another Context subject is selected for validation using a balanced schedule with seed 10, yielding 462–465 training, 35–37 validation, and 35–37 test windows. The training and validation sets are combined to form the Context set.

MAMI [56]

Data Source and Content MAMI contains over 10,000 image–text memes annotated for misogyny and fine-grained misogyny categories. We use the binary misogyny-detection task.

Link https://github.com/MIND-Lab/SemEval2022-Task5-Multimedia-Automatic-Misogyny-Identification-MAMI

Feature Extraction Images are encoded using a frozen DINOv3 ViT-B/16, and text is encoded using Qwen3-Embedding-0.6B.

Data Splits The official split is used, containing 9,000 training, 1,000 validation, and 1,000 test memes. The random seed is fixed at 42.

PAD-UFES-20 [57]

Data Source and Content PAD-UFES-20 contains smartphone clinical images and patient metadata for six-class skin-lesion diagnosis.

Link https://data.mendeley.com/datasets/zr7vgbcyr2

Feature Extraction Clinical images are encoded using a frozen ImageNet-pretrained ResNet-50, while patient metadata is retained as a structured modality using identity mapping.

Data Splits A patient-grouped stratified five-fold split with seed 42 is used. Each fold contains 1,838–1,839 Context and 459–460 Test samples, and results are averaged over the five folds.

B More Experimental Details

B.1 Supplementary Results for the Main Experiment

Table 3 provides the complete quantitative results corresponding to the main experiment. For all specialized fusion methods, the reported results are averaged over three independent training runs with different random seeds $(1, 2, 3)$, as these methods require task-specific parameter optimization. In contrast, our method performs prediction through parameter-free multimodal in-context inference without adaptation. Therefore, its results are deterministic under the same evaluation setting and are reported directly without variance estimation.

Table 3: Comparison results across different multimodal datasets. Results of the specialized fusion methods are reported as mean ± standard deviation, with the standard deviation shown below the mean in the same secondary-line font size as the ▲/▼ marker. The marker below each Ours ACC or AUC reports its increase/decrease relative to the mean of the corresponding metric over the specialized fusion methods.

Dataset	Concat		EFT		MTO		CAF		TMC		**Ours**	
	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑
GAMMA	65.00 (± 7.07)	**85.63** (± 2.31)	58.33 (± 8.50)	76.11 (± 16.66)	65.00 (± 7.07)	79.07 (± 4.81)	60.00 (± 4.08)	81.89 (± 1.81)	63.33 (± 4.71)	76.30 (± 7.20)	**70.00** (▲ 7.67)	83.33 (▲ 3.53)
MUStARD	**73.67** (± 0.34)	**81.17** (± 1.37)	70.05 (± 1.90)	74.37 (± 2.10)	68.84 (± 2.13)	75.02 (± 0.29)	71.98 (± 0.34)	78.10 (± 0.92)	70.29 (± 1.18)	78.37 (± 0.22)	70.29 (▼ 0.68)	77.83 (▲ 0.42)
TCGA-KIRC	65.66 (± 2.86)	63.09 (± 2.53)	**74.75** (± 2.86)	**77.13** (± 3.41)	70.71 (± 11.43)	65.84 (± 10.68)	67.68 (± 3.78)	71.21 (± 4.03)	61.62 (± 1.43)	62.74 (± 5.52)	72.73 (▲ 4.65)	71.49 (▲ 3.49)
TCGA-LUAD	52.46 (± 1.34)	57.17 (± 0.92)	**60.66** (± 2.68)	**62.42** (± 3.14)	59.02 (± 1.34)	58.33 (± 4.73)	56.83 (± 3.86)	61.15 (± 1.90)	50.82 (± 8.78)	51.16 (± 9.06)	**60.66** (▲ 4.70)	45.16 (▼ 12.89)
NYUDv2	56.17 (± 0.79)	84.13 (± 0.72)	40.06 (± 3.42)	68.09 (± 5.05)	45.06 (± 2.45)	69.71 (± 4.10)	53.41 (± 1.55)	82.13 (± 2.27)	53.16 (± 1.08)	81.24 (± 1.12)	**59.33** (▲ 9.76)	**87.40** (▲ 10.34)
N-MNIST+N-TIDIGITS	**95.64** (± 0.08)	**99.79** (± 0.05)	58.88 (± 3.92)	92.71 (± 0.61)	94.17 (± 0.91)	99.64 (± 0.14)	94.93 (± 0.13)	99.78 (± 0.04)	92.27 (± 1.74)	99.32 (± 0.35)	94.28 (▲ 7.10)	99.70 (▲ 1.45)
MultiOFF	56.82 (± 3.21)	55.73 (± 7.26)	59.06 (± 1.64)	57.39 (± 1.78)	60.18 (± 2.47)	61.78 (± 1.80)	58.17 (± 2.28)	58.25 (± 2.34)	62.19 (± 1.76)	64.25 (± 1.38)	**63.76** (▲ 4.48)	**65.80** (▲ 6.32)
MET-Meme(E)	34.91 (± 1.18)	67.33 (± 2.73)	25.75 (± 2.63)	56.78 (± 1.54)	33.18 (± 3.07)	61.06 (± 3.03)	34.60 (± 2.79)	63.30 (± 3.91)	36.97 (± 2.54)	**68.58** (± 1.97)	**38.39** (▲ 5.31)	66.02 (▲ 2.61)
RAVDESS	55.00 (± 0.88)	89.43 (± 0.71)	23.54 (± 3.73)	68.80 (± 1.68)	**63.75** (± 1.84)	87.95 (± 2.79)	57.64 (± 1.49)	88.39 (± 0.56)	51.74 (± 2.99)	86.95 (± 0.77)	60.00 (▲ 9.67)	**92.51** (▲ 8.21)
Handwritten	96.67 (± 0.59)	99.55 (± 0.08)	38.33 (± 3.03)	88.15 (± 0.65)	89.83 (± 3.77)	98.89 (± 0.68)	97.83 (± 0.42)	99.77 (± 0.06)	94.75 (± 0.82)	98.00 (± 0.64)	**98.75** (▲ 15.27)	**99.89** (▲ 3.02)
Memotion	**78.28** (± 0.07)	57.53 (± 0.47)	77.65 (± 0.56)	54.13 (± 3.25)	78.14 (± 0.18)	56.49 (± 2.43)	78.14 (± 0.07)	**58.16** (± 0.49)	77.06 (± 1.03)	55.05 (± 2.24)	78.18 (▲ 0.33)	56.30 (▲ 0.03)
SIIM-ISIC	**97.83** (± 0.00)	72.78 (± 3.40)	97.81 (± 0.03)	80.31 (± 4.97)	**97.83** (± 0.00)	74.78 (± 8.99)	97.78 (± 0.05)	74.72 (± 4.43)	**97.83** (± 0.00)	64.59 (± 19.17)	**97.83** (▲ 0.01)	**81.13** (▲ 7.69)
Twitter1517	**76.65** (± 0.27)	**63.77** (± 0.29)	75.44 (± 0.93)	51.40 (± 4.30)	76.08 (± 0.27)	62.58 (± 3.04)	75.94 (± 0.53)	62.23 (± 1.01)	74.62 (± 0.74)	62.07 (± 1.42)	76.36 (▲ 0.61)	63.32 (▲ 2.91)
WESAD	54.47 (± 1.96)	62.90 (± 2.11)	72.52 (± 0.66)	76.94 (± 2.70)	71.84 (± 0.34)	82.38 (± 1.82)	55.30 (± 2.25)	70.21 (± 5.66)	61.80 (± 6.38)	76.85 (± 7.92)	**83.35** (▲ 20.16)	**93.35** (▲ 19.49)
Enrico	51.60 (± 0.58)	**85.81** (± 1.09)	31.85 (± 2.84)	67.53 (± 2.51)	46.35 (± 1.86)	74.70 (± 2.92)	50.11 (± 0.32)	84.41 (± 0.72)	**52.28** (± 1.43)	84.98 (± 1.28)	50.68 (▲ 4.24)	83.51 (▲ 4.02)
MELD	53.84 (± 1.12)	68.65 (± 1.82)	53.23 (± 1.36)	64.39 (± 2.86)	62.83 (± 1.79)	**76.61** (± 1.24)	61.62 (± 0.57)	76.26 (± 0.18)	56.77 (± 2.76)	74.22 (± 0.90)	**64.29** (▲ 6.63)	76.49 (▲ 4.46)
PAD-UFES-20	75.01 (± 0.75)	91.98 (± 0.07)	68.80 (± 1.21)	79.81 (± 1.44)	73.05 (± 0.48)	87.96 (± 1.48)	72.96 (± 0.24)	90.65 (± 0.17)	72.92 (± 0.29)	90.05 (± 0.44)	**76.85** (▲ 4.30)	**93.04** (▲ 4.95)
Derm7pt	63.12 (± 0.66)	83.73 (± 0.24)	49.96 (± 7.43)	67.84 (± 1.68)	61.10 (± 1.06)	78.93 (± 1.46)	65.49 (± 1.21)	86.72 (± 1.07)	55.53 (± 0.48)	77.11 (± 0.88)	**72.41** (▲ 13.37)	**90.00** (▲ 11.13)

B.2 Supplementary Results for the Effectiveness of Multimodal Extension

Table 4 provides the complete comparison between TabPFN and our model across diverse real-world datasets. Overall, our model achieves stronger performance across a broad range of multimodal prediction tasks, with improvements observed over heterogeneous modality combinations. These results demonstrate that the proposed model effectively broadens the model's capability to capture and utilize predictive correlations among heterogeneous modalities.

Table 4: Comparison of TabPFN and our model across diverse datasets.

Dataset	TabPFN		Ours	
	ACC↑	AUC↑	ACC↑	AUC↑
GAMMA	65.00	80.67	**70.00**	**83.33**
MUStARD	**71.74**	**79.92**	70.29	77.83
TCGA-KIRC	69.70	69.83	**72.73**	**71.49**
TCGA-LUAD	54.10	**58.11**	**60.66**	45.16
NYUDv2	59.02	**89.17**	**59.33**	87.40
N-MNIST+N-TIDIGITS	**95.92**	**99.78**	94.28	99.70
Enrico	49.66	80.75	**50.68**	**83.51**
MultiOFF	**63.76**	64.30	**63.76**	**65.80**
MET-Meme(E)	35.55	64.16	**38.39**	**66.02**
RAVDESS	55.21	88.25	**60.00**	**92.51**
Handwritten	**98.75**	**99.94**	**98.75**	99.89
Memotion	**78.18**	53.46	**78.18**	**56.30**
SIIM-ISIC	**97.83**	**84.14**	**97.83**	81.13
Twitter1517	**76.47**	**65.55**	76.36	63.32
WESAD	79.34	90.90	**83.35**	**93.35**
MELD	**67.01**	**78.97**	64.29	76.49
PAD-UFES-20	76.29	**93.16**	**76.85**	93.04
Twitter2015	**67.31**	**81.68**	65.48	78.53